

Program -10-Documentation

Title: Demonstrate Kubernetes Dashboard using CLI and Kubernetes Cluster

Objective: Deploy an Nginx application, expose it via a service, and monitor the infrastructure using the Kubernetes Dashboard.

1. Project Initialization

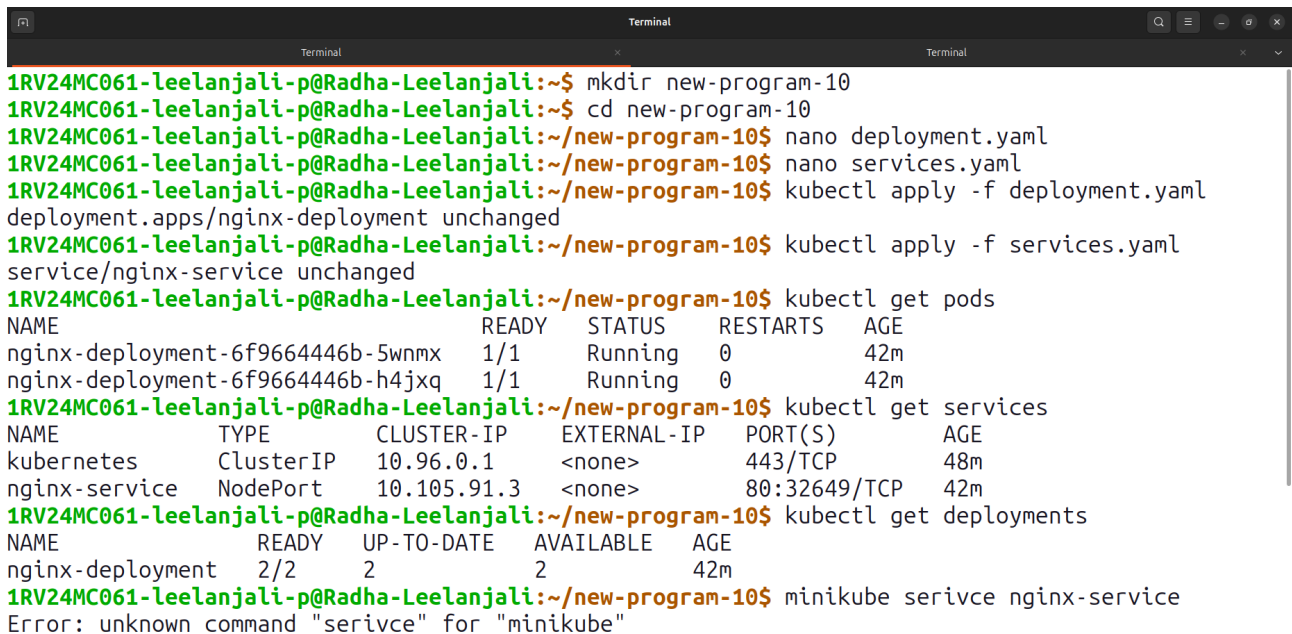
Before applying configurations, create a dedicated workspace to keep your YAML files organized.

- `mkdir new-program-10`: Creates a new directory for the project.
- `cd new-program-10`: Moves the terminal context into the project folder.

2. Configuration Creation

The nano command is used here as a text editor to define the desired state of your cluster.

- `nano deployment.yaml`: Used to define the **Deployment** (specifying the Nginx image, number of replicas, and labels).
- `nano services.yaml`: Used to define the **Service** (specifying how to expose the pods to external traffic using a NodePort).



```
1RV24MC061-lee@lee:~$ mkdir new-program-10
1RV24MC061-lee@lee:~/new-program-10$ cd new-program-10
1RV24MC061-lee@lee:~/new-program-10$ nano deployment.yaml
1RV24MC061-lee@lee:~/new-program-10$ nano services.yaml
1RV24MC061-lee@lee:~/new-program-10$ kubectl apply -f deployment.yaml
deployment.apps/nginx-deployment unchanged
1RV24MC061-lee@lee:~/new-program-10$ kubectl apply -f services.yaml
service/nginx-service unchanged
1RV24MC061-lee@lee:~/new-program-10$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-5wnmx   1/1     Running   0           42m
nginx-deployment-6f9664446b-h4jxq   1/1     Running   0           42m
1RV24MC061-lee@lee:~/new-program-10$ kubectl get services
NAME              TYPE          CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
kubernetes        ClusterIP     10.96.0.1    <none>         443/TCP           48m
nginx-service     NodePort      10.105.91.3  <none>         80:32649/TCP     42m
1RV24MC061-lee@lee:~/new-program-10$ kubectl get deployments
NAME              READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment  2/2     2             2           42m
1RV24MC061-lee@lee:~/new-program-10$ minikube service nginx-service
Error: unknown command "service" for "minikube"
```

Did you mean this?
service

3. Applying Configurations

The kubectl apply command sends your YAML definitions to the Kubernetes API server.

- **Command:** kubectl apply -f deployment.yaml
 - *Purpose:* Creates the Pods based on the deployment strategy. In your case, it resulted in 2 running Nginx pods.
- **Command:** kubectl apply -f services.yaml
 - *Purpose:* Creates a logical shim (Service) that routes traffic to pods labeled with app: nginx.

4. Verification and Status Checks

It is essential to verify that resources are "Running" and "Available."

Command	Purpose
kubectl get pods	Lists all active pods. Shows status (e.g., Running), readiness, and age.
kubectl get services	Displays the Service Type (NodePort), Cluster-IP, and the assigned Port (e.g., 32649).
kubectl get deployments	Shows the high-level status of the deployment, confirming if the desired number of replicas are updated and available.

```
Terminal
kubernetes      ClusterIP  10.96.0.1    <none>      443/TCP      48m
nginx-service   NodePort   10.105.91.3  <none>      80:32649/TCP 42m
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/new-program-10$ kubectl get deployments
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment  2/2     2            2           42m
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/new-program-10$ minikube service nginx-service
Error: unknown command "service" for "minikube"

Did you mean this?
    service

Run 'minikube --help' for usage.
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/new-program-10$ minikube service nginx-service

```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-service	80	http://192.168.49.2:32649

```
🐞 Opening service default/nginx-service in default browser...
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/new-program-10$ ln: failed to create symbolic link '/home/1RV24MC061-leelanjali-p/snap/firefox/7477/.config/dconf/user': File exists
Gtk-Message: 15:41:44.588: Not loading module "atk-bridge": The functionality is provided by GTK natively. Please try to not load it.
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/new-program-10$
```

```
Terminal
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/new-program-10$ minikube dashboard
🤖 Verifying dashboard health ...
🚀 Launching proxy ...
😞 Verifying proxy health ...
🎉 Opening http://127.0.0.1:38489/api/v1/namespaces/kubernetes-dashboard/services/http:kubern
es-dashboard:/proxy/ in your default browser...
ln: failed to create symbolic link '/home/1RV24MC061-leelanjali-p/snap/firefox/7477/.config/dcon
f/user': File exists
Gtk-Message: 15:41:56.936: Not loading module "atk-bridge": The functionality is provided by GTK
natively. Please try to not load it.
```

5. Accessing the Application

Since you are using **Minikube** (a local Kubernetes tool), the service isn't automatically reachable via a standard public IP.

- **Command:** minikube service nginx-service
 - *Action:* This generates a local URL (e.g., `http://192.168.49.2:32649`) and automatically opens your default web browser to the "Welcome to nginx!" page.

6. Cluster Visualization

To see a graphical representation of your labels, namespaces, and resource health:

- **Command:** minikube dashboard
 - *Action:* Starts a local proxy and opens the **Kubernetes Dashboard** in your browser.
 - *Insight:* As seen in your screenshot, this provides a "Workload Status" overview where you can see your 1 Deployment, 2 Pods, and 1 ReplicaSet represented by green health charts.

